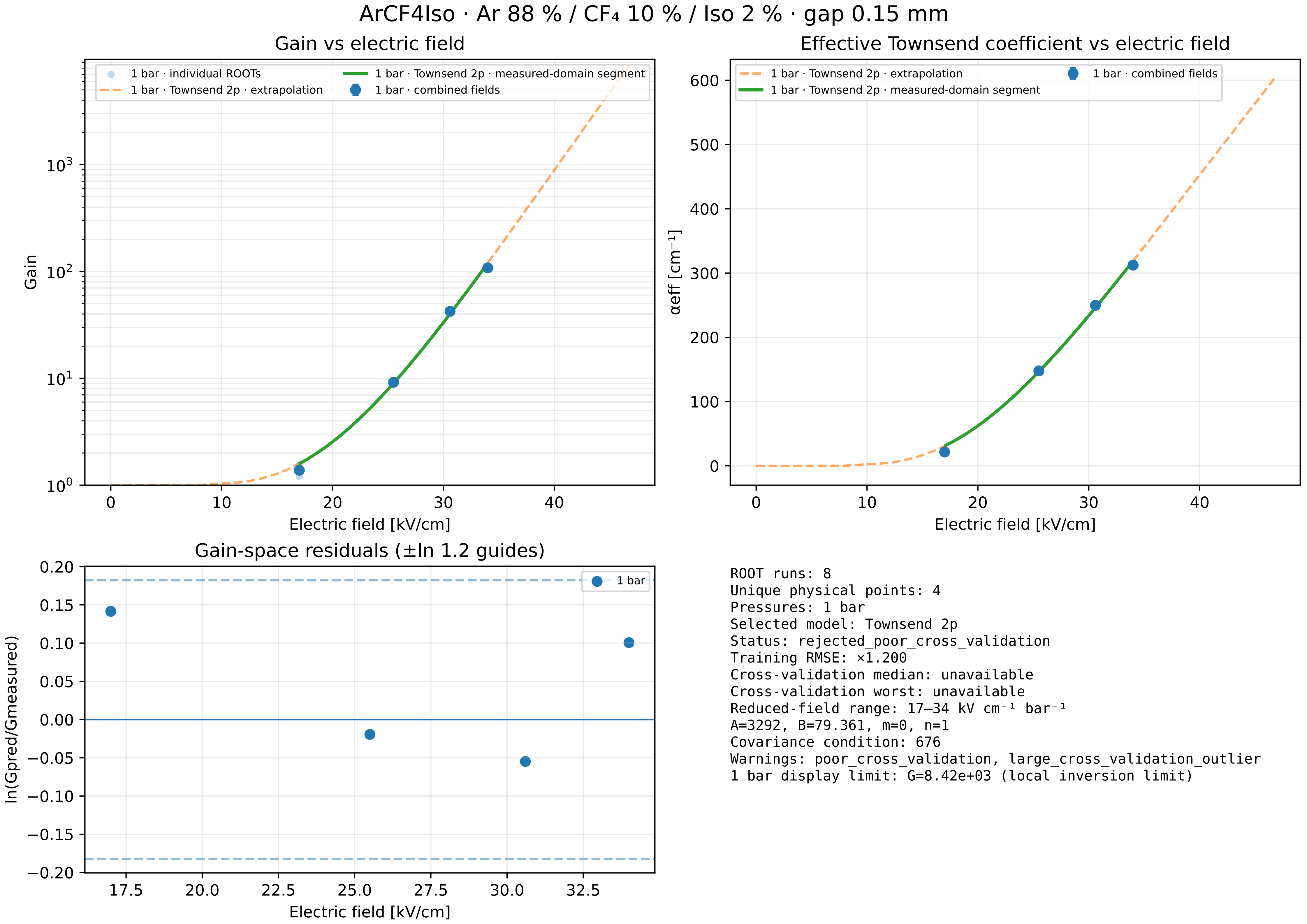
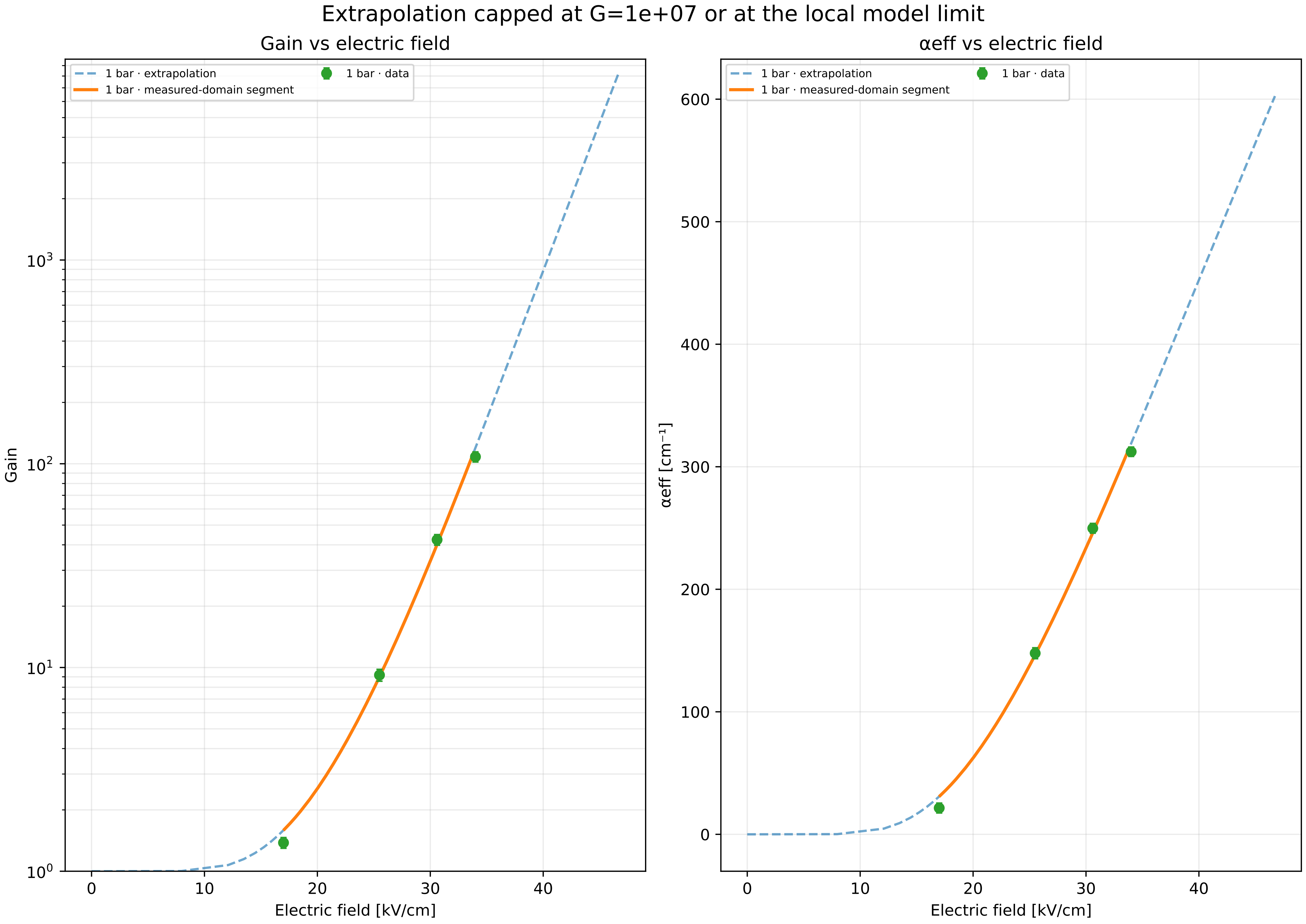

ArCF4Iso · Ar 88 % / CF₄ 10 % / Iso 2 % · gap 0.15 mm
Gain vs electric field
1 bar · individual ROOTs
1 bar · Townsend 2p · extrapolation
1 bar · Townsend 2p · measured-domain segment
1 bar · combined fields
Gain
Electric field [kV/cm]
Effective Townsend coefficient vs electric field
αeff [cm⁻¹]
Electric field [kV/cm]
Gain-space residuals (±ln 1.2 guides)
1 bar
ln(Gpred/Gmeasured)
Electric field [kV/cm]
ROOT runs: 8
Unique physical points: 4
Pressures: 1 bar
Selected model: Townsend 2p
Status: rejected_poor_cross_validation
Training RMSE: ×1.200
Cross-validation median: unavailable
Cross-validation worst: unavailable
Reduced-field range: 17–34 kV cm⁻¹ bar⁻¹
A=3292, B=79.361, m=0, n=1
Covariance condition: 676
Warnings: poor_cross_validation, large_cross_validation_outlier
1 bar display limit: G=8.42e+03 (local inversion limit)



Extrapolation capped at G=1e+07 or at the local model limit
Gain vs electric field
1 bar · extrapolation
1 bar · measured-domain segment
1 bar · data
Gain
10^0
10^1
10^2
10^3
0
10
20
30
40
Electric field [kV/cm]
αeff vs electric field
1 bar · extrapolation
1 bar · measured-domain segment
1 bar · data
αeff [cm⁻¹]
0
100
200
300
400
500
600
0
10
20
30
40
Electric field [kV/cm]

Exact combined points used by the fit · page 1/1
Repeated ROOTs at the same field are pooled once; all source ROOTs remain on disk.

p [bar]	E [kV/cm]	E/p	Gain	σGain	αeff [cm⁻¹]	npe	runs
1	17	17	1.38	0.0834	21.4722	200	2
1	25.5	25.5	9.18	0.613	147.802	200	2
1	30.6	30.6	42.37	2.54	249.763	200	2
1	34	34	108.2	6.2	312.265	200	2

Progressive model selection
The higher-order model is selected only when new points improve out-of-sample prediction.

Candidate	k	AICc	CV median	CV worst	usable	warnings
Townsend 2p	2	2.4	—	—	no	poor cross validation, large cross validation outlier